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HEAD-UP DISPLAY

Abstract

A head-up display includes: a second housing including a wall; and a plate-shaped cover supported by a placement surface of the housing (second housing) and covering a second opening of the housing. The wall includes: a pair of first extensions extending in the X-axis direction; a pair of second extensions extending in the Y-axis direction; and a plurality of claw parts disposed along the wall. The cover includes a plurality of first engagement portions formed corresponding to the plurality of claw parts. The plurality of claw parts include pressers protruding from the wall toward the second opening. The second housing holds the cover in a curved state by surrounding the cover with the pair of first extensions and the pair of second extensions and sandwiching the cover between the placing surface and the pressers engaged with the plurality of first engagement portions.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION
[0001] The present application is based on and claims priority of Japanese Patent Application No. 2024-028278 filed on Feb. 28, 2024.
FIELD
[0002] The present disclosure relates to a head-up display.
BACKGROUND
[0003] Patent Literature (PTL) 1 discloses a projection device provided in a vehicle and capable of projecting an image. The projection device includes a housing that includes a wall provided with an opening for allowing display light to pass toward the outside, and a plate-shaped cover that is translucent and closes the opening from the outside. The wall includes a pair of edges extending in a first direction, and a plurality of claw parts arranged in the first direction on each of the pair of edges. The cover includes a plurality of through holes arranged corresponding to the plurality of claw parts. The claw part includes a base protruding outward from the edge, and a presser protruding in the first direction from the leading end of the base. The housing holds the cover in a state where the cover is sandwiched between the edge and the pressers inserted into the through holes.
CITATION LIST
Patent Literature
[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2022-123194
SUMMARY
[0005] However, the projection device according to PTL 1 can be improved upon.
[0006] Therefore, the present disclosure provides a head-up display capable of improving upon the above related art.
[0007] A head-up display according to one aspect of the present disclosure is a head-up display provided in a vehicle to project image light onto an object capable of transmitting and reflecting light, and form a virtual image on a side opposite to an observer relative to the object, the head-up display including: an image-generating device configured to project the image light onto the object; a housing that houses the image-generating device, and includes an opening and a wall disposed around the opening, the opening allowing the image light to pass to outside; and a cover that is plate-shaped, is supported by a placement surface of the housing, and covers the opening. The wall includes: a pair of first extensions facing each other across the opening and extending in a first direction; a pair of second extensions extending in a second direction different from the first direction; and a plurality of claw parts arranged along the wall. The pair of first extensions are curved toward the image-generating device. The cover includes a plurality of first engagement portions that are provided corresponding to the plurality of claw parts. Each of the plurality of claw parts includes a presser that protrudes from the wall toward the opening. The housing holds the cover in a curved state by surrounding the cover with the pair of first extensions and the pair of second extensions and sandwiching the cover between the placement surface and the presser engaged with each of the plurality of first engagement portions.
[0008] According to the head-up display of the present disclosure, it is possible to inhibit the cover from lifting while inhibiting an increase in size.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0010] FIG. **1** is a perspective view illustrating a head-up display, provided in a vehicle, according to an embodiment.

[0011] FIG. **2** is a perspective view illustrating the head-up display according to the embodiment.

[0012] FIG. **3** is an exploded perspective view illustrating a cover and a second housing of the head-up display according to the embodiment.

[0013] FIG. **4** is a sectional view illustrating a windshield and the head-up display.

[0014] FIG. **5** is a partial plan view illustrating a part of the second housing, including a wall, a placement surface, a plurality of claw parts, protrusions, and the like.

[0015] FIG. **6** is a perspective view illustrating a first claw part and a first engagement portion.

[0016] FIG. **7** is a sectional view illustrating the first claw part and the first engagement portion along line A**2**-A**2** in FIG. **6**.

[0017] FIG. **8** is a sectional view illustrating the cover, the wall of the second housing, and the drainage structure of the second housing, along line A**1**-A**1** in FIG. **2**.

[0018] FIG. **9** is an explanatory view illustrating how the cover is attached to the second housing.

DESCRIPTION OF EMBODIMENT

[0019] Hereinafter, an exemplary embodiment will be specifically described with reference to the drawings.

[0020] Note that the embodiment described below shows either general or specific examples. Numerical values, shapes, components, placement positions and connection forms of the components, and the like shown in the following embodiment are examples and are not intended to limit the present disclosure. Among the components in the following embodiment, components not recited in the independent claim will be described as optional components.

[0021] Each of the drawings is a schematic diagram and is not necessarily illustrated exactly. Therefore, for example, the scale and the like may not necessarily be consistent across the drawings. In the drawings, the same reference numerals are assigned to substantially the same components, and a duplicate description may be omitted or simplified.

[0022] In the following embodiment, in FIG. **2**, the front side of the vehicle is defined as the positive X-axis direction, and the right side, when viewed along the front of the vehicle, is defined as the positive Y-axis direction. The upper side in the vertical direction, perpendicular to the positive X-axis and the positive Y-axis directions, is defined as the positive Z-axis direction. The direction of FIG. **2** is also applied in FIG. **3** and subsequent figures.

[0023] In the following embodiment, expressions such as X-axis direction, rectangular shape, and substantially parallel are used. For example, the X-axis direction does not necessarily mean the perfect X-axis direction, the rectangular shape does not necessarily mean perfectly rectangular, and substantially parallel does not necessarily mean perfectly parallel. However, these expressions also mean substantially the X-axis direction, substantially rectangular, and substantially parallel, that is, including a margin of error, such as a few percent. In addition, the X-axis direction, rectangular shape, and substantially parallel mean the X-axis direction, rectangular shape, and substantially parallel in the range where the effects of the present disclosure can be achieved. The same applies to other expressions using “direction”, “shape”, or “substantially”.

Embodiment

Configuration

[0024] Head-up display **1** of the present disclosure will be specifically described below with reference to FIGS. **1** to **9**.

[0025] FIG. **1** is a perspective view illustrating head-up display **1**, provided in vehicle **2**, according to an embodiment. FIG. **2** is a perspective view illustrating head-up display **1** according to the embodiment. FIG. **3** is an exploded perspective view illustrating cover **40** and second housing **20** of head-up display **1** according to the embodiment. FIG. **4** is a sectional view illustrating windshield **3** and head-up display **1**. FIG. **5** is a partial plan view illustrating a part of second housing **20**, including wall **21**, placement surface **120**, the plurality of claw parts **24**, protrusions **25**, and the like. FIG. **6** is a perspective view illustrating first claw part **24***a* and first engagement portion **41**. FIG. **7** is a sectional view illustrating first claw part **24***a* and first engagement portion **41** along line A**2**-A**2** in FIG. **6**. FIG. **8** is a sectional view illustrating cover **40**, wall **21** of second housing **20**, and drainage structure **26** of second housing **20**, along line A**1**-A**1** in FIG. **2**. FIG. **9** is an explanatory view illustrating how cover **40** is attached to second housing **20**. (a) of FIG. **9** illustrates a case where claw part **24** is engaged with first engagement portion **41**, protrusion **25** is engaged with second engagement portion **42**, and cover **40** is placed on placement surface **120** of second housing **20**. (b) of FIG. **9** is a sectional view illustrated along line B-B in (a) of FIG. **9**. (c) of FIG. **9** is a sectional view illustrated along line C-C in (a) of FIG. **9**. (d) of FIG. **9** illustrates a state where cover **40** is slid in the positive Y-axis direction from the state in (a) of FIG. **9**, and cover **40** is sandwiched between the plurality of claw parts **24** and placement surface **120** of second housing **20**. (e) of FIG. **9** is a sectional view illustrated along line E-E in (d) of FIG. **9**. (f) of FIG. **9** is a sectional view illustrated along line F-F in (d) of FIG. **9**.

[0026] As illustrated in FIG. **1**, head-up display **1** is disposed, for example, on dashboard **5** (also referred to as an instrument panel) of vehicle **2**. Windshield **3** (also referred to as a front shield) is disposed above dashboard **5** of vehicle **2**. Head-up display **1** is disposed in dashboard **5**.

[0027] Head-up display **1** is provided in vehicle **2**, projects an image onto an object capable of transmitting and reflecting light, and forms a virtual image on the side opposite to an observer relative to the object. For example, head-up display **1** can reflect image light, representing an image emitted from image-generating device **50**, onto windshield **3** as an object to display a virtual image represented by the image light for a user such as a driver. That is, head-up display **1** projects the image light emitted from image-generating device **50** onto the front of windshield **3** to display an image indicated by the image light on windshield **3**. The image projected onto windshield **3** of vehicle **2** is thus visible by the user.

[0028] Here, the image light is light representing information of an image including numbers, characters, figures, and the like, and is displayed as a virtual image in the front of windshield **3**. The image is a still image or a moving image, and is an image of numbers, characters, figures, and the like.

[0029] As illustrated in FIGS. **2** to **4**, head-up display **1** includes first housing **10**, image-generating device **50**, first reflective mirror **31**, second reflective mirror **32**, second housing **20**, and cover **40**.

[0030] First housing **10** is an enclosure that houses image-generating device **50**, first reflective mirror **31**, and second reflective mirror **32**. First housing **10** constitutes the outer shell of head-up display **1**. First housing **10** is fixed to vehicle **2** while attached to dashboard **5**. First housing **10** is made of resin material or metal material.

[0031] First opening **11** for disposing second housing **20** is formed on the positive Z-axis direction side of first housing **10**. First opening **11** is provided to correspond to the shape and position of second opening **22** formed in second housing **20**.

[0032] Second housing **20** is attached to first housing **10**. Specifically, an engagement portion, disposed at the terminal edge of second housing **20** on the negative Z-axis direction side and extending in the negative Z-axis direction, is located at the terminal edge of first housing **10** on the positive Z-axis direction side and engaged with an engaged portion extending in the positive Z-axis direction. A mating structure is formed in which the engagement portion of second housing **20** is located outside the engaged portion of first housing **10**. This makes it difficult for liquid to enter the interior of head-up display **1**.

[0033] Image-generating device **50** generates an image and outputs image light indicating the generated image so that the image is projected onto windshield **3**. More specifically, image-generating device **50** emits image light from a display component. The image light emitted from the display component of image-generating device **50** is reflected by first reflective mirror **31** and then also reflected by second reflective mirror **32**, passes through first opening **11**, and enters and transmits through cover **40**. The image light then passes through second opening **22** and is emitted from head-up display **1** to be projected onto windshield **3**. That is, image-generating device **50** can project a predetermined image onto windshield **3** by emitting image light. When the image light is reflected by windshield **3**, the user can recognize a virtual image. Image-generating device **50** is, for example, a liquid crystal display device including a liquid crystal display and other components.

[0034] Image-generating device **50** includes a display component, a container, a light-emitting module, and a condenser lens.

[0035] The display component is a liquid crystal display element such as a liquid crystal display. In the display component, the light emitted from the light-emitting module is emitted from the back surface side of the display component so that the emitting surface emits light. In the display component, image light, indicating an image including numbers, characters, figures, and the like, is emitted from the emitting surface in response to a control instruction from an electronic control unit (ECU) installed in vehicle **2**.

The display component is driven by AC power obtained from a power supply in vehicle **2**.

[0036] The container is an enclosure that houses a display component, a light-emitting module, and a condenser lens. The container is disposed on the bottom plate of first housing **10**.

[0037] The light-emitting module includes a light source and a substrate on which the light source is mounted. The light source is mounted on the substrate that is held to emit light toward the display component from the back side of the display component through the condenser lens. The light source is formed by, for example, light-emitting diodes. For example, the light source is driven by DC power obtained from the power supply in vehicle **2**.

[0038] The condenser lens is disposed on the light emission direction side of the light source and disposed on an optical path between the light source and the display component. The condenser lens emits light emitted from the light source toward the display component.

[0039] First reflective mirror **31** reflects the image light emitted from image-generating device **50** toward second reflective mirror **32**. Second reflective mirror **32** reflects the image light emitted from image-generating device **50** toward cover **40** through first opening **11**.

[0040] Specifically, first reflective mirror **31** and second reflective mirror **32** are arranged in first housing **10** so as to face each other while being separated by a predetermined distance. More specifically, first reflective mirror **31** is disposed on the negative X-axis direction side of first housing **10** so as to face the emitting surface of image-generating device **50** and second reflective mirror **32**, which are arranged on the positive X-axis direction side of first housing **10**. First reflective mirror **31** and second reflective mirror **32** are arranged between image-generating device **50** and first opening **11**. Second reflective mirror **32** faces first reflective mirror **31** and is disposed on the positive X-axis direction side of first reflective mirror **31**. That is, second reflective mirror **32** is disposed to be aligned with first reflective mirror **31** in the X-axis direction.

[0041] First reflective mirror **31** and second reflective mirror **32** are concave mirrors, convex mirrors, or plane mirrors. In the present embodiment, each of first reflective mirror **31** and second reflective mirror **32** is a concave mirror with a free-form surface. In the present embodiment, each of first reflective mirror **31** and second reflective mirror **32** is a rectangular mirror that is long in the Y-axis direction. Note that the shapes of first reflective mirror **31** and second reflective mirror **32** are not particularly limited, and may be polygonal or circular.

[0042] Second reflective mirror **32** may be swingable about the Y-axis direction. In this case, the position of the image projected onto windshield **3** can be adjusted. Second reflective mirror **32** may be swingable by manual operation or electric operation using a drive mechanism.

[0043] Second housing **20** is disposed closer to windshield **3** than first housing **10** and coupled to first housing **10**. Second housing **20** is made of resin material. A structure that includes first housing **10** and second housing **20** is an example of a housing.

[0044] As illustrated in FIG. **3**, second housing **20** has a flat rectangular shape and is a frame-shaped body with rectangular second opening **22**, formed in a center portion, for allowing image light to pass to the outside. Note that the shape of second housing **20** and the shape of second opening **22** are not limited to a rectangular shape. The shape of second housing **20** and the shape of second opening **22** may be set as appropriate according to the space for installing head-up display **1** in vehicle **2**. Therefore, the shape of second housing **20** and the shape of second opening **22** may be a polygonal shape other than a rectangular shape, a circular shape, or any other known shape.

[0045] Second housing **20** supports plate-shaped cover **40** covering second opening **22**. Placement surface **120** for placing cover **40** is formed on second housing **20**. With cover **40** placed and supported on placement surface **120**, cover **40** covers second opening **22**. As illustrated in FIG. **4**, second opening **22** faces first opening **11** of first housing **10**, and second opening **22** and first opening **11** are arranged side by side in the Z-axis direction.

[0046] As illustrated in FIGS. **3** and **4**, second housing **20** includes wall **21** disposed around second opening **22**. Specifically, second opening **22** is surrounded by placement surface **120**, and wall **21** is disposed around placement surface **120**.

[0047] Placement surface **120** is curved so that the center portion in the X-axis direction is recessed. Specifically, placement surface **120** includes a pair of first placement surfaces **121** extending in the X-axis direction and a pair of second placement surfaces **122** extending in the Y-axis direction. The pair of first placement surfaces **121** are inclined upward in the negative X-axis direction and curved so that the inclination of tangent lines tangent to the pair of first placement surfaces **121** with respect to the X-axis direction gradually increases. The Y-axis direction of the pair of second placement surfaces **122** is substantially parallel to the Y-axis direction, but may be curved to be recessed in the negative Z-axis direction. In the present embodiment, the pair of first placement surfaces **121** and the pair of second placement surfaces **122** may be collectively referred to as placement surface **120**.

[0048] Wall **21** rises from the surface of second housing **20** on the positive Z-axis direction side, that is, from the terminal edge of placement surface **120**, and surrounds second opening **22** and placement surface **120**. Surrounding second opening **22** and placement surface **120** may refer to completely surrounding second opening **22** and placement surface **120**, or substantially surrounding these with one or more drainage structures **26** formed and cut into wall **21**.

[0049] The outer peripheral shape of wall **21** surrounding second opening **22** is asymmetric in the first direction and the second direction (X-axis direction and Y-axis direction) and is similar to the shape of second opening **22**. Although wall **21** of the present embodiment does not completely surround second opening **22** with drainage structure **26** or the like, the outer peripheral shape of wall **21** is a shape regarded as completely surrounding second opening **22**. The X-axis direction is an example of the first direction. The Y-axis direction is an example of the second direction.

[0050] Wall **21** includes a pair of first extensions **21***a* facing each other across second opening **22** and extending in the X-axis direction, a pair of second extensions **21***b* extending in the Y-axis direction in a direction different from the X-axis direction, and a plurality of claw parts **24** arranged along wall **21**.

[0051] The pair of first extensions **21***a* extend in the X-axis direction. One first extension **21***a* of the pair of first extensions **21***a* is disposed on the positive Y-axis direction side of second opening **22**. The other first extension **21***a* of the pair of first extensions **21***a* is disposed on the negative Y-axis direction side of second opening **22**.

[0052] As described above, first placement surface **121** is curved, so that the pair of first extensions **21***a* is curved toward image-generating device **50**, as illustrated in FIG. **4**.

[0053] As illustrated in FIG. **3**, the pair of second extensions **21***b* extend in the Y-axis direction. One second extension **21***b***1** of the pair of second extensions **21***b* is disposed on the front side (positive X-axis direction side) of vehicle **2**. Other second extension **21***b***2** of the pair of second extensions **21***b* is disposed on the rear side (negative X-axis direction side) of vehicle **2**.
[0054] As illustrated in FIG. **5**, one second extension **21***b***1** is longer than other second extension **21***b***2**. Note that one second extension **21***b***1** and other second extension **21***b***2** may have the same length. Other second extension **21***b***2** may be longer than one second extension **21***b***1**.
[0055] In the present embodiment, the interval between two adjacent claw parts **24** provided on one second extension **21***b***1** is smaller than the interval between two adjacent claw parts **24** provided on other second extension **21***b***2**, and one second extension **21***b***1** is longer than other second extension **21***b***2**. Therefore, the number of claw parts **24** provided on one second extension **21***b***1** is smaller than the number of claw parts **24** provided on other second extension **21***b***2**.
[0056] Each of the plurality of claw parts **24** includes a presser protruding from wall **21** toward second opening **22**. The plurality of claw parts **24** include first claw parts **24***a* arranged along the pair of first extensions **21***a* and second claw parts **24***b* arranged along the pair of second extensions **21***b*.
[0057] Specifically, first claw parts **24***a* are arranged along the pair of first extensions **21***a*. In the present embodiment, two or more first claw parts **24***a* are arranged side by side along the pair of first extensions **21***a*. Two or more first claw parts **24***a* are provided on the pair of first placement surfaces **121** formed along the pair of first extensions **21***a*. In the present embodiment, three first claw parts **24***a* are provided on each of the pair of first placement surfaces **121**. Note that the number of first claw parts **24***a* provided on each of the pair of first placement surfaces **121** may be two or less, or four or more. The number of first claw parts **24***a* provided on one first placement surface **121** of the pair of first placement surfaces **121** and the number of first claw parts **24***a* provided on the other first placement surface **121** of the pair of first placement surfaces **121** may be the same or different.
[0058] As illustrated in FIGS. **6** and **7**, first claw part **24***a* includes base **225** protruding in the positive Z-axis direction and first presser **224***a* protruding in the negative X-axis direction from the leading end of base **225**. First presser **224***a* also extends in the Y-axis direction toward second opening **22**. That is, first presser **224***a* extends from base **225** in the Y-axis direction and in the negative X-axis direction. First presser **224***a* is an example of a presser.
[0059] Since first presser **224***a* faces first placement surface **121** with cover **40** therebetween, the movement of cover **40** in the Z-axis direction can be inhibited.
[0060] First presser **224***a* is a protruding plate extending from each of the pair of first extensions **21***a* toward second opening **22**. Inclined portion **224***b* inclined upward away from first placement surface **121** is coupled to first presser **224***a*. Inclined portion **224***b* is also coupled to base **225**.
[0061] Inclined portion **224***b* is a plate portion inclined upward in the negative Y-axis direction. Inclined portion **224***b* extends from base **225** in the negative X-axis direction and extends from first presser **224***a* in the negative Y-axis direction. Inclined portion **224***b* can guide cover **40** to first presser **224***a* when cover **40** is attached to second housing **20**.
[0062] As illustrated in FIG. **8**, drainage structure **26** is formed in second housing **20**. Drainage structure **26** is a drainage hole penetrating in a direction orthogonal to the longitudinal direction of the pair of first extensions **21***a* or a recess in the direction orthogonal to the longitudinal direction of the pair of first extensions **21***a*. Drainage structure **26** extends vertically downward (in the negative Z-axis direction) from the surface of cover **40**. Thus, even if water drops fall on the surface of cover **40**, the water drops are drained from the drainage structure **26**.
[0063] Drainage structures **26** are formed at positions corresponding to the plurality of first claw parts **24***a* arranged along each of the pair of first extensions **21***a*. That is, drainage structures **26** are formed in the negative Z-axis direction of the plurality of first claw parts **24***a*.
[0064] As illustrated in FIG. **5** and (a) of FIG. **9**, second claw part **24***b* protrudes in the X-axis

direction from each of the pair of second extensions **21***b* toward second opening **22**. The second claw part **24***b* includes second presser **224***c* extending from each of the pair of second extensions **21***b* toward second opening **22**, and inclined portion **224***d* coupled to second presser **224***c* and each of the pair of second extensions **21***b* and inclined upward away from second placement surface **122** with respect to second presser **224***c*. Second presser **224***c* is an example of a presser. In the present embodiment, first presser **224***a* and second presser **224***c* may be collectively referred to as a presser.

[0065] Second presser **224***c* disposed on one second extension **21***b***1** extends in the negative X-axis direction from the surface of one second extension **21***b***1** on the side of second opening **22**. Second presser **224***c* disposed on other second extension **21***b***2** extends in the positive X-axis direction from the surface of other second extension **21***b***2** on the side of second opening **22**. Since second presser **224***c* faces second placement surface **122** with cover **40** therebetween, the movement of cover **40** in the Z-axis direction can be inhibited.

[0066] Second presser **224***c* is a protruding plate extending from each of the pair of second extensions **21***b* toward second opening **22**. Inclined portion **224***d* inclined upward away from second placement surface **122** is coupled to second presser **224***c*.

[0067] Inclined portion **224***d* of second extension **21***b* is a plate portion inclined upward in the negative Y-axis direction. Inclined portion **224***d* can guide cover **40** when cover **40** is attached to second housing **20**.

[0068] Second placement surface **122** includes protrusions **25** disposed along the pair of second extensions **21***b*. Protrusions **25** engage with cover **40** to inhibit the movement of cover **40** in the X-axis direction and the Y-axis direction.

[0069] Protrusions **25** are located on both end sides and the center side of one second extension **21***b***1** and at least on the center side of other second extension **21***b***2**. Protrusions **25** may further be located on both end sides of other second extension **21***b***2**. Protrusion **25** located on the center side of one second extension **21***b***1** and protrusion **25** located on the center side of other second extension **21***b***2** do not overlap in the X-axis direction. That is, protrusion **25** located on the center side of one second extension **21***b***1** and protrusion **25** located on the center side of other second extension **21***b***2** are displaced in the Y-axis direction so as not to overlap in the X-axis direction.

[0070] Drainage structure **26** is formed in one second extension **21***b***1**. Drainage structure **26** is a drainage hole penetrating in a direction orthogonal to the longitudinal direction of one second extension **21***b***1**, or a notched recess cut in a direction orthogonal to the longitudinal direction of other second extension **21***b***1**. Drainage structure **26** extends vertically downward (in the negative Z-axis direction) from the surface of cover **40**. Thus, even if water drops fall on the surface of cover **40**, the water drops are drained from the drainage structure **26**. Note that drainage structure **26** may also be formed in other second extension **21***b***2**.

[0071] Drainage structures **26** are formed at positions corresponding to the plurality of second claw parts **24***b* arranged along each of the pair of second extensions **21***b*. That is, drainage structures **26** are formed in the negative Z-axis direction of the plurality of second claw parts **24***b*.

[0072] Cover **40** is made of a translucent resin material such as polycarbonate (PC) or acrylic, or a polarizing plate. Since the image light transmitted through first opening **11** passes through second opening **22**, the image light transmitted through cover **40** is projected onto windshield **3**.

[0073] Cover **40** also serves as an antifouling cover for inhibiting dust from entering first housing **10**. For example, the surface of cover **40** on the side of windshield **3** may be subjected to hard coating treatment. In this case, even if dust adheres to the surface, dirt can be easily wiped off. The surface of cover **40** on the side of windshield **3** is the surface of cover **40** on the positive X-axis direction side, and is an emitting surface from which image light is emitted. The surface of cover **40** may have an antireflection coating.

[0074] Cover **40** is supported by placement surface **120** of second housing **20**, wall **21** of second housing **20**, the plurality of claw parts **24**, and the plurality of protrusions **25**.

[0075] Specifically, cover **40** includes a plurality of first engagement portions **41** formed corresponding to the plurality of claw parts **24**, and a plurality of second engagement portions **42** formed corresponding to the plurality of protrusions **25**. In the present embodiment, the plurality of first engagement portions **41** are at least either a plurality of through holes or a plurality of notches.
[0076] The plurality of first engagement portions **41** are formed at positions corresponding to the plurality of claw parts **24** when cover **40** is placed on placement surface **120** of second housing **20**, and the plurality of second engagement portions **42** are formed at positions corresponding to the plurality of protrusions **25** when cover **40** is placed on placement surface **120** of the housing. Second engagement portion **42** of cover **40** is a long hole in the Y-axis direction that allows cover **40** to slide.
[0077] As illustrated in FIG. **6**, first engagement portion **41** disposed on the negative Y-axis direction side includes the following: first engagement hole **41***a*, through which claw part **24** provided on first placement surface **121** is inserted when cover **40** is assembled to second housing **20**; and second engagement hole **41***b*, through which base **225** is inserted when cover **40** is slid in the positive Y-axis direction. First engagement portion **41** disposed on the negative Y-axis direction side has an L-shape formed by first engagement hole **41***a* and second engagement hole **41***b*. As illustrated in FIG. **7**, when base **225** is inserted in second engagement hole **41***b*, cover **40** is placed to be sandwiched between first presser **224***a* and first placement surface **121**.
[0078] Cover **40** has a shape corresponding to the outer peripheral shape of wall **21**. Therefore, the shape of cover **40** is asymmetric in the X-axis direction and the Y-axis direction and is similar to the shape of second opening **22**.
[0079] In second housing **20**, cover **40** is surrounded by the pair of first extensions **21***a* and the pair of second extensions **21***b* by wall **21** and second engagement portions **42** of cover **40**. Cover **40** can be sandwiched between the pressers of the plurality of claw parts **24** inserted into first engagement portions **41** and placement surface **120** of second housing **20**, using the plurality of claw parts **24** and the plurality of first engagement portions **41**. The plurality of protrusions **25** engage with second engagement portions **42** in a state where cover **40** is held by the pressers, the pair of first extensions **21***a*, and the pair of second extensions **21***b*. That is, second housing **20** can be engaged with cover **40** using the plurality of protrusions **25** and the plurality of second engagement portions **42** of cover **40**. As a result, second housing **20** can hold cover **40** in a curved state. That is, second housing **20** can hold cover **40** in a curved state along placement surface **120**, which is curved so that the center portion in the X-axis direction is recessed.
[0080] Therefore, even if cover **40** is pressed when, for example, dust adhering to the surface of cover **40** is wiped off, cover **40** can be inhibited from lifting because cover **40** is supported by wall **21**, the plurality of claw parts **24**, and the plurality of protrusions **25**.
[0081] In head-up display **1**, image light emitted from image-generating device **50** enters first reflective mirror **31** and is reflected by first reflective mirror **31**. The image light then enters second reflective mirror **32**, is reflected by second reflective mirror **32**, passes through first opening **11**, and enters cover **40** supported by second opening **22** of second housing **20**. The image light entering cover **40** transmits cover **40**, and the image light is projected onto windshield **3**.
[0082] A procedure for attaching cover **40** to second housing **20** in head-up display **1** will be described with reference to (a) to (f) of FIG. **9**.
[0083] (a) to (f) of FIG. **9** are schematic explanatory views illustrating how cover **40** is attached to second housing **20**.
[0084] As illustrated in (a) to (c) of FIG. **9**, cover **40** is moved in the negative Z-axis direction to cover second opening **22** of second housing **20**.
[0085] Specifically, cover **40** is placed on placement surface **120** to cover second opening **22** of second housing **20** so that the plurality of claw parts **24** of second housing **20** and the plurality of first engagement portions **41** of cover **40** correspond one-to-one, and the plurality of protrusions **25** of second housing **20** and the plurality of second engagement portions **42** of cover **40** correspond

one-to-one.

[0086] Cover **40** is engaged with second housing **20** while being lowered in the negative Z-axis direction so that the plurality of claw parts **24** of second housing **20** and the plurality of first engagement portions **41** of cover **40** correspond one-to-one, and the plurality of protrusions **25** of second housing **20** and the plurality of second engagement portions **42** of cover **40** correspond one-to-one. Thus, the plurality of claw parts **24** of second housing **20** and the plurality of first engagement portions **41** of cover **40** are engaged one-to-one, and the plurality of protrusions **25** of second housing **20** and the plurality of second engagement portions **42** of cover **40** are engaged one-to-one. As a result, cover **40** is placed on placement surface **120** of second housing **20**.

[0087] Next, cover **40** is slid in the positive Y-axis direction. As illustrated in (d) to (f) of FIG. **9**, the sliding of cover **40** moves the plurality of first engagement portions **41**, and cover **40** is placed between placement surface **120** and the pressers of the plurality of claw parts **24**. In the present embodiment, the sliding cover **40** in the positive Y-axis direction enables cover **40** to be sandwiched between placement surface **120** and the pressers of the plurality of claw parts **24** inserted into the plurality of first engagement portions **41**.

[0088] Thus, cover **40** is inhibited from moving in the Z-axis direction by the pressers of the plurality of claw parts **24** and inhibited from moving in the X-axis direction and the positive Y-axis direction by the plurality of claw parts **24** and the plurality of protrusions **25**. Therefore, second housing **20** can be inhibited from lifting against placement surface **120**.

Operations and Effects

[0089] Next, the operations and effects of head-up display **1** according to the present embodiment will be described.

[0090] For example, in a head-up display, when a cover is attached to a housing, the cover may be attached to the housing using a separate member such as double-sided tape or pins. In this case, the installation cost and the assembling man-hour increase due to the necessity for jigs and pressurizing equipment, and the material cost of the separate member is required. This results in higher cost, as well as a larger outside shape of the wall of the head-up display with an opening, which may affect the ease of in-vehicle mounting and the appearance design of a dashboard (also referred to as an instrument panel).

[0091] In contrast, the projection device of PTL 1 discloses the configuration in which the housing holds the cover in a state where the cover is sandwiched between the edge and the pressers inserted into the through holes. However, when the cover is pressed from the outside or continuously exposed to a high-temperature environment, the cover may deform and lift from the edge. In this case, there is a problem that the quality of the cover deteriorates, and a gap is formed between the cover and the edge, allowing dust to enter through the gap.

[0092] Therefore, as described above, head-up display **1** according to technique **1** of the present embodiment is head-up display **1** provided in vehicle **2** to project image light onto an object capable of transmitting and reflecting light, and form a virtual image on the side opposite to an observer relative to the object. Head-up display **1** includes: image-generating device **50** configured to project the image light onto the object; a housing (second housing **20**) that houses image-generating device **50** and includes an opening (second opening **22**) for allowing the image light to pass to the outside, and wall **21** disposed around the opening (second opening **22**); and plate-shaped cover **40** supported by placement surface **120** of the housing (second housing **20**), and covering the opening (second opening **22**). Wall **21** includes: a pair of first extensions **21***a* facing each other across the opening (second opening **22**) and extending in a first direction (X-axis direction); a pair of second extensions **21***b* extending in a second direction (Y-axis direction) that is a direction different from the first direction (X-axis direction); and a plurality of claw parts **24** arranged along wall **21**. The pair of first extensions **21***a* are curved toward image-generating device **50**. Cover **40** includes a plurality of first engagement portions **41** that are formed corresponding to the plurality of claw parts **24**. Each of the plurality of claw parts **24** includes a presser that protrudes from wall **21** toward the opening (second opening **22**). The housing (second housing **20**) holds cover **40** in a curved state by surrounding cover **40** with the pair of first extensions **21***a* and the pair of second extensions **21***b* and sandwiching cover **40** between placement surface **120** and the pressers engaged with the plurality of first engagement portions **41**.

[0093] With this configuration, cover **40** can be sandwiched between the plurality of claw parts **24** and placement surface **120**, and thus, even if cover **40** supported by placement surface **120** is pressed, wall **21** and the plurality of claw parts **24** can support cover **40**. Therefore, the plurality of claw parts **24** can inhibit cover **40** from lifting.

[0094] In addition, since cover **40** is sandwiched between the plurality of claw parts **24** and placement surface **120**, it is not necessary to separately couple cover **40** to second housing **20** using a double-sided tape or the like.

[0095] Therefore, according to the present disclosure, it is possible to inhibit cover **40** from lifting while inhibiting an increase in size. According to the present disclosure, since cover **40** can be held by second housing **20** without using double-sided tape or the like, substantial increases in the material cost and manufacturing cost of head-up display **1** can be inhibited.

[0096] Further, head-up display **1** according to technique **2** of the present embodiment is head-up display **1** according to technique **1**, wherein the plurality of claw parts **24** are provided on the pair of second extensions **21***b*, and the presser protrudes in the first direction (X-axis direction) toward the opening (second opening **22**).

[0097] With this configuration, the lifting of cover **40** placed along the pair of second extensions **21***b* can be inhibited. In particular, the lifting of cover **40** in the front-rear direction of vehicle **2** can be inhibited.

[0098] Further, head-up display **1** according to technique **3** of the present embodiment is head-up display **1** according to technique **2**, wherein the plurality of claw parts **24** are arranged along the pair of first extensions **21***a*, and the presser extends in the second direction (Y-axis direction) toward the opening (second opening **22**).

[0099] With this configuration, the lifting of cover **40** placed along the pair of first extensions **21***a* can be inhibited. In particular, the lifting of cover **40** in the left-right direction of vehicle **2** can be inhibited.

[0100] Further, head-up display **1** according to technique **4** of the present embodiment is head-up display **1** according to technique **3**, wherein one second extension **21***b***1** of the pair of second extensions **21***b* is disposed on the front side of vehicle **2**, other second extension **21***b***2** of the pair of second extensions **21***b* is disposed on the rear side of vehicle **2**, and the number of claw parts **24** provided on one second extension **21***b***1** is smaller than the number of claw parts **24** provided on other second extension **21***b***2**.

[0101] With this configuration, the lifting from one second extension **21***b***1** can be further inhibited. For example, when one second extension **21***b***1** is longer than other second extension **21***b***2**, the number of claw parts **24** provided on one second extension **21***b***1** can be increased, so that the lifting from one second extension **21***b***1** can be inhibited.

[0102] It is considered that in particular, when the surface of cover **40** in head-up display **1** is pressed from the seat side, cover **40** is often pressed from the rear of vehicle **2** toward the front of vehicle **2**. In this case, pressing from the rear to the front of vehicle **2** may cause the front side of cover **40** to lift off first placement surface **121**. However, according to the present embodiment, the lifting from one second extension **21***b***1** can be further inhibited, thus inhibiting dust from entering through the gap between second housing **20** and cover **40**.

[0103] Further, head-up display **1** according to technique **5** of the present embodiment is head-up display **1** according to any one of techniques **1** to **4**, wherein cover **40** includes second engagement portion **42**, placement surface **120** (second placement surface **122**) includes protrusion **25** disposed along each of the pair of second extensions **21***b*, and protrusion **25** is engaged with second engagement portion **42** in a state where cover **40** is held by the presser, the pair of first extensions

21_a_, and the pair of second extensions **21**_b_.

[0104] With this configuration, when cover **40** is attached to second housing **20**, claw part **24** and first engagement portion **41** of cover **40** correspond to each other, and protrusion **25** of second housing **20** and second engagement portion **42** of cover **40** correspond to each other, so that cover **40** can be slid in a predetermined direction without rattling. That is, the engagement of claw part **24** with first engagement portion **41** and the engagement of protrusion **25** with second engagement portion **42** make it possible to position cover **40** and inhibit the movement of cover **40** in the X-axis direction. As a result, cover **40** can be easily attached to second housing **20**.

[0105] Further, head-up display **1** according to technique **6** of the present embodiment is head-up display **1** according to technique **5**, wherein one second extension **21**_b_**1** of the pair of second extensions **21**_b_ is disposed on the front side of vehicle **2**, other second extension **21**_b_**2** of the pair of second extensions **21**_b_ is disposed on the rear side of vehicle **2**, and protrusions **25** are located on both end sides and the center side of one second extension **21**_b_**1**, and located at least on the center side of other second extension **21**_b_**2**.

[0106] With this configuration, the lifting from one second extension **21**_b_**1** can be further inhibited. For example, when one second extension **21**_b_**1** is longer than other second extension **21**_b_**2**, the number of claw parts **24** provided on one second extension **21**_b_**1** can be increased, so that the lifting from one second extension **21**_b_**1** can be inhibited.

[0107] Further, head-up display **1** according to technique **7** of the present embodiment is head-up display **1** according to technique **6**, wherein protrusions **25** are further located on both end sides of other second extension **21**_b_**2**.

[0108] With this configuration, the lifting from one second extension **21**_b_**1** can be further inhibited.

[0109] Further, head-up display **1** according to technique **8** of the present embodiment is head-up display **1** according to technique **6** or **7**, wherein the outer peripheral shape of wall **21** surrounding the opening (second opening **22**) is asymmetric in the X-axis direction and the Y-axis direction and is similar to the shape of the opening (second opening **22**), and protrusion **25** located on the center side of one second extension **21**_b_**1** and protrusion **25** located on the center side of other second extension **21**_b_**2** do not overlap in the X-axis direction.

[0110] With this configuration, since protrusions **25** can be disposed on the center side of one second extension **21**_b_**1** and the center side of other second extension **21**_b_**2** according to the outer peripheral shape of the asymmetric wall **21**, even if these protrusions **25** do not overlap in the X-axis direction, wall **21**, the plurality of claw parts **24**, and the plurality of protrusions **25** can support cover **40**. Therefore, the lifting of cover **40** can be inhibited.

[0111] Further, head-up display **1** according to technique **9** of the present embodiment is head-up display **1** according to any one of techniques **1** to **8**, wherein one second extension **21**_b_**1** of the pair of second extensions **21**_b_ is disposed on the front side of vehicle **2**, other second extension **21**_b_**2** of the pair of second extensions **21**_b_ is disposed on the rear side of vehicle **2**, and one second extension **21**_b_**1** is longer than other second extension **21**_b_**2**.

[0112] With this configuration, the shape of head-up display **1** can be set in accordance with the size and shape of the space of vehicle **2** in which head-up display **1** is disposed.

[0113] Further, head-up display **1** according to technique **10** of the present embodiment is head-up display **1** according to any one of techniques **1** to **9**, wherein drainage structure **26** is formed in each of the pair of first extensions **21**_a_, and drainage structure **26** extends vertically downward from the surface of cover **40**.

[0114] With this configuration, even if the liquid adheres to the surface of cover **40**, the liquid on the surface of cover **40** can be drained out of drainage structure **26**. Therefore, the image light entering cover **40** can be properly projected onto windshield **3** without the liquid entering the interior of head-up display **1**.

[0115] Drainage structure **26** formed in each of the pair of first extensions **21**_a_ is larger than drainage structure **26** formed in each of the pair of second extensions **21**_b_. This not only drains the liquid adhering to the surface of cover **40**, but also makes it easier for the operator's fingers to enter and perform assembly, positioning, and sliding operations. Of three drainage structures **26** formed in each of the pair of first extensions **21***a*, all three drainage structures **26** may be larger than drainage structure **26** formed in the pair of second extensions **21***b*, or one or two drainage structures **26** of three drainage structures **26** may be larger than drainage structure **26** formed in the pair of second extensions **21***b*.

[0116] Further, head-up display **1** according to technique **11** of the present embodiment is head-up display **1** according to any one of techniques **1** to **10**, wherein one second extension **21***b***1** of the pair of second extensions **21***b* is disposed on the front side of vehicle **2**, drainage structure **26** is formed in one second extension **21***b***1**, and drainage structure **26** extends vertically downward from the surface of cover **40**.

[0117] With this configuration, even if liquid adheres to the surface of cover **40**, the liquid flows to one second extension **21***b***1** in the vertical downward direction, allowing the liquid on the surface of cover **40** to drain out of drainage structure **26** of one second extension **21***b***1**. Therefore, the image light entering cover **40** can be properly projected onto windshield **3** without the liquid entering the interior of head-up display **1**.

[0118] Further, head-up display **1** according to technique **12** of the present embodiment is head-up display **1** according to any one of techniques **1** to **11**, wherein the presser (first presser **224***a*) extends from each of the pair of first extensions **21***a* toward the opening (second opening **22**), and inclined portion **224***b* inclined upward away from placement surface **120** is coupled to the presser (first presser **224***a*).

[0119] With this configuration, during the attachment of cover **40** to second housing **20**, inclined portion **224***b* can guide cover **40** as cover **40** is placed on placement surface **120** and slid in the Y-axis direction. Therefore, cover **40** can be sandwiched between the plurality of claw parts **24** and placement surface **120**.

[0120] Further, head-up display **1** according to technique **13** of the present embodiment is head-up display **1** according to any one of techniques **1** to **12**, wherein the presser (second presser **224***c*) extends from each of the pair of second extensions **21***b* toward the opening (second opening **22**), and inclined portion **224***d* inclined upward away from placement surface **120** is coupled to the presser (second presser **224***c*).

[0121] With this configuration, during the attachment of cover **40** to second housing **20**, inclined portion **224***d* can guide cover **40** as cover **40** is placed on placement surface **120** and slid in the Y-axis direction. Therefore, cover **40** can be sandwiched between the plurality of claw parts **24** and placement surface **120**.

Other Variations, Etc.

[0122] Although the head-up display according to the present disclosure has been described based on the embodiment described above, the present disclosure is not limited to the embodiment. Ones obtained by applying various modifications conceivable by those skilled in the art to the embodiment may also be included within the scope of the present disclosure, as long as they do not depart from the gist of the present disclosure.

[0123] Note that the present disclosure also includes forms obtained by applying various modifications conceivable by those skilled in the art to the above embodiment, and forms realized by arbitrarily combining components and functions in the embodiment without departing from the gist of the present disclosure.

Further Information about Technical Background to this Application

[0124] The disclosure of the following patent application including specification, drawings, and claims is incorporated herein by reference in their entirety: Japanese Patent Application No. 2024-028278 filed on Feb. 28, 2024.

INDUSTRIAL APPLICABILITY

[0125] The present disclosure is applicable to mobile objects such as vehicles, for example.

Claims

1. A head-up display provided in a vehicle to project image light onto an object capable of transmitting and reflecting light, and form a virtual image on a side opposite to an observer relative to the object, the head-up display comprising: an image-generating device configured to project the image light onto the object; a housing that houses the image-generating device, and includes an opening and a wall disposed around the opening, the opening allowing the image light to pass to outside; and a cover that is plate-shaped, is supported by a placement surface of the housing, and covers the opening, wherein the wall includes: a pair of first extensions facing each other across the opening and extending in a first direction; a pair of second extensions extending in a second direction different from the first direction; and a plurality of claw parts arranged along the wall, the pair of first extensions are curved toward the image-generating device, the cover includes a plurality of first engagement portions that are provided corresponding to the plurality of claw parts, each of the plurality of claw parts includes a presser that protrudes from the wall toward the opening, and the housing holds the cover in a curved state by surrounding the cover with the pair of first extensions and the pair of second extensions and sandwiching the cover between the placement surface and the presser engaged with each of the plurality of first engagement portions.
2. The head-up display according to claim 1, wherein the plurality of claw parts are provided on the pair of second extensions, and the presser protrudes in the first direction toward the opening.
3. The head-up display according to claim 2, wherein the plurality of claw parts are arranged along the pair of first extensions, and the presser extends in the second direction toward the opening.
4. The head-up display according to claim 3, wherein one second extension of the pair of second extensions is disposed on a front side of the vehicle, an other second extension of the pair of second extensions is disposed on a rear side of the vehicle, and a total number of claw parts provided on the one second extension is smaller than a total number of claw parts provided on the other second extension.
5. The head-up display according to claim 1, wherein the cover includes a second engagement portion, the placement surface includes a protrusion disposed along each of the pair of second extensions, and the protrusion is engaged with the second engagement portion in a state where the cover is held by the presser, the pair of first extensions, and the pair of second extensions.
6. The head-up display according to claim 5, wherein one second extension of the pair of second extensions is disposed on the front side of the vehicle, an other second extension of the pair of second extensions is disposed on the rear side of the vehicle, and the protrusion is located on each of a center side and both end sides of the one second extension, and is located at least on a center side of the other second extension.
7. The head-up display according to claim 6, wherein the protrusion is further located on each of both end sides of the other second extension.
8. The head-up display according to claim 6, wherein an outer peripheral shape of the wall surrounding the opening is asymmetric in the first direction and the second direction and is similar to a shape of the opening, and the protrusion located on the center side of the one second extension and the protrusion located on the center side of the other second extension do not overlap in the first direction.
9. The head-up display according to claim 1, wherein one second extension of the pair of second extensions is disposed on the front side of the vehicle, an other second extension of the pair of second extensions is disposed on the rear side of the vehicle, and the one second extension is longer than the other second extension.
10. The head-up display according to claim 1, wherein a drainage structure is provided on each of the pair of first extensions, and the drainage structure extends vertically downward from a surface of the cover.

11. The head-up display according to claim 1, wherein one second extension of the pair of second extensions is disposed on the front side of the vehicle, a drainage structure is provided on the one second extension, and the drainage structure extends vertically downward from a surface of the cover.

12. The head-up display according to claim 1, wherein the presser extends from each of the pair of first extensions toward the openings, and an inclined portion inclined upward away from the placement surface is coupled to the presser.

13. The head-up display according to claim 1, wherein the pressers extend from each of the pair of second extensions toward the openings, and an inclined portion inclined upward away from the placement surface is coupled to the presser.